

Netflix Movie Recommendation System using Dynamic Hybrid SVD

1.Problem Understanding

Modern streaming platforms contain thousands of movies, making content discovery increasingly difficult for users. Recommendation systems help solve this problem by learning user preferences from historical interactions and suggesting relevant content.

In this project, a personalized movie recommendation system was developed using the Netflix Prize dataset. Multiple collaborative filtering approaches, including User-Based Collaborative Filtering, Item-Based Collaborative Filtering, and Matrix Factorization (SVD), were implemented and compared. The final system uses a Dynamic Hybrid SVD approach that combines latent factor modeling with popularity-based weighting to improve recommendations for cold-start users.

The objectives of this project are:

- Develop a personalized movie recommendation engine.
 - Compare multiple recommendation algorithms.
 - Evaluate rating prediction performance using RMSE.
 - Evaluate ranking quality using MAP@10.
 - Address cold-start user scenarios.
 - Develop a Streamlit-based interactive dashboard for recommendation generation.
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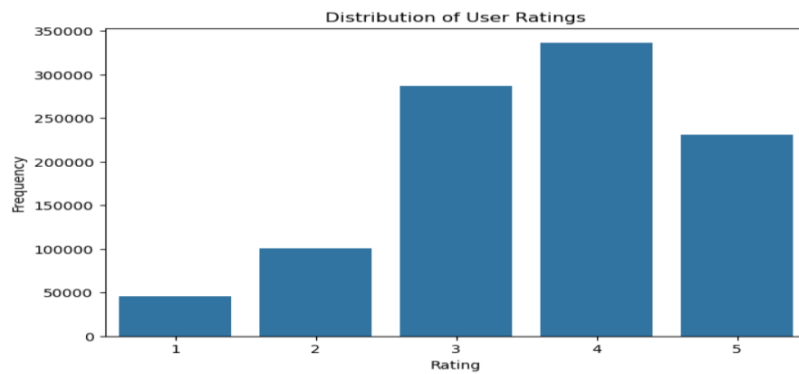
2. Exploratory Data Analysis

2.1 Data Preparation

The original Netflix Prize dataset contains over 100 million ratings. To enable efficient experimentation and model training, the first 1 million rating records were selected. Movies with missing release year information were removed during preprocessing, and duplicate records were checked and eliminated.

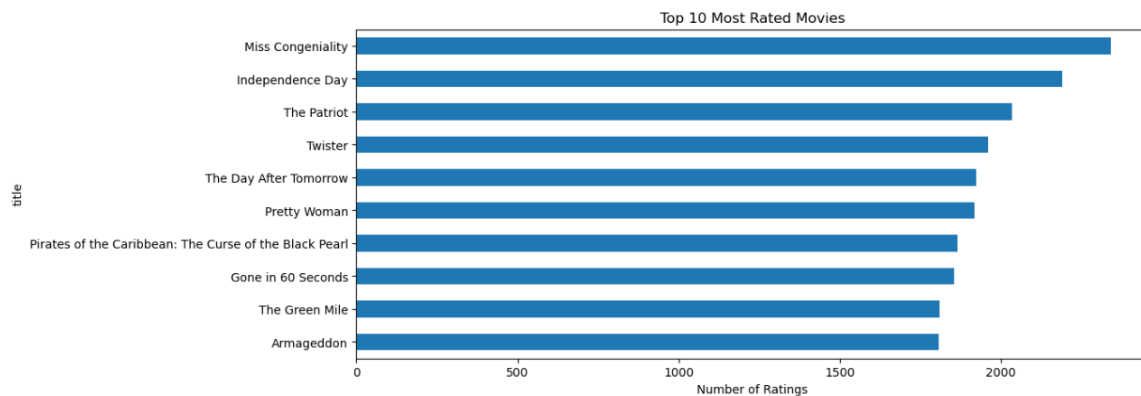
Metric	Value
Total Ratings	999,994
Unique Users	289,387
Unique Movies	16,144
Average Rating	3.61

2.2 Rating Distribution



Most ratings fall between 3 and 5 stars, indicating generally positive user sentiment. Lower ratings occur less frequently, resulting in a positively skewed distribution.

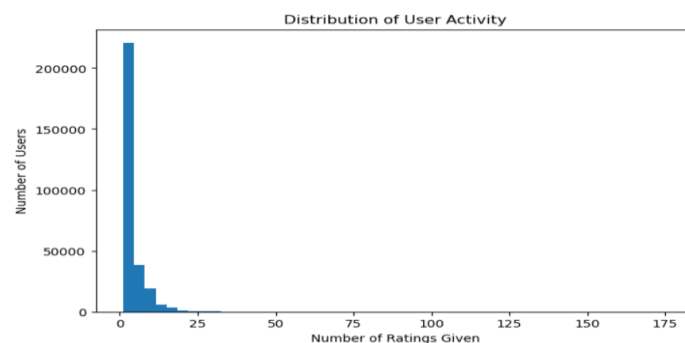
2.3 Movie Popularity Analysis



The most frequently rated movies are mainstream and commercially successful titles such as *“Miss Congeniality”*, *“Independence Day”* and *“The Patriot”*.

The distribution suggests that user interactions are concentrated around a relatively small set of highly popular movies. This phenomenon is common in recommendation systems and contributes to the long-tail nature of user-item interactions.

2.4 User Activity Analysis



User activity is highly uneven. Most users rate only a few movies, while a small number of users contribute hundreds of ratings. This creates challenges for recommendation systems, particularly for cold-start users.

2.5 Data Sparsity Analysis

Dataset Sparsity = 99.9786%

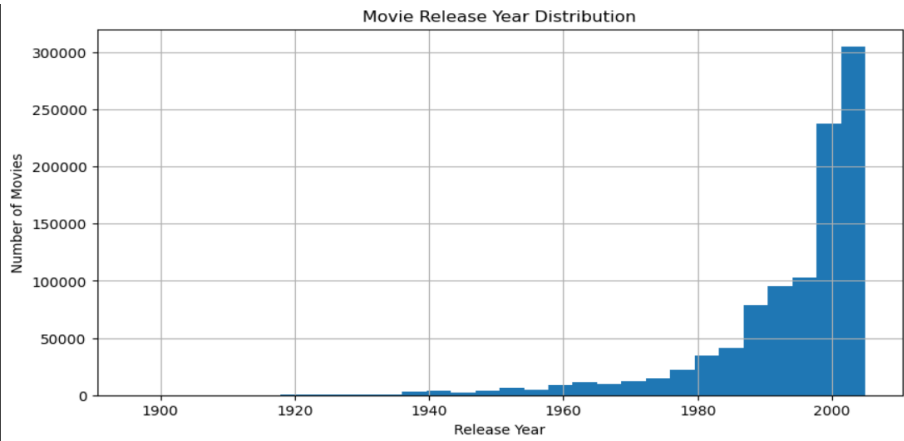
The user-item matrix is highly sparse because users interact with only a small fraction of available movies. This sparsity motivates the use of Matrix Factorization (SVD), which performs well in sparse recommendation environments.

2.6 Cold User Analysis

Metric	Value
Cold Users (≤ 3 Ratings)	196,941
Percentage of Users	68.05%

A significant number of users have very limited rating histories. To improve recommendations for such users, dynamic popularity weighting was incorporated into the final recommendation system.

2.7 Movie Release Year Distribution



Most movies were released between 1990 and 2005, indicating that the dataset is dominated by relatively recent content. However, movies from earlier decades are also present, providing diversity in content age and enabling recommendations across multiple generations of films.

These observations motivated the selection of SVD as the primary recommendation model and the use of dynamic popularity weighting to improve recommendations for users with limited interaction history.

3. Methodology

Three collaborative filtering approaches were implemented and evaluated for movie recommendation:

3.1 User-Based Collaborative Filtering (UserCF)

- Finds users with similar rating patterns.
- Predicts ratings using preferences of similar users.
- Effective for personalization but sensitive to sparse data.

3.2 Item-Based Collaborative Filtering (ItemCF)

- Identifies movies that receive similar ratings.
- Recommends movies related to items previously liked by the user.
- More stable than UserCF but still affected by sparsity.

3.3 Matrix Factorization (SVD)

- Decomposes the user-item matrix into latent user and movie factors.
- Learns hidden relationships between users and movies.
- Predicts ratings for unseen movies using latent factor interactions.

Model Selection

All three approaches were evaluated using RMSE and MAP@10. SVD achieved the best performance and was selected as the foundation of the final recommendation system. The selected model was further enhanced using dynamic popularity weighting to improve recommendations for cold-start users.

4. Model Design

The final recommendation system is based on a Dynamic Hybrid SVD architecture that combines Matrix Factorization with popularity-based ranking. The objective was to improve recommendation quality while addressing cold-start user scenarios.

4.1 SVD Model Configuration

The recommendation engine uses Singular Value Decomposition (SVD) from the Surprise library. The final model was trained using the following hyperparameters:

Parameter	Value
n_factors	200

Parameter	Value
n_epochs	50
learning_rate	0.005
regularization	0.02

These parameters were selected to improve the model's ability to learn latent user preferences while maintaining good generalization performance.

4.2 Popularity Score

In addition to predicted ratings, a popularity score was computed for each movie.

Popularity Score:

Popularity Score = Average Rating $\times$ $\log(1 + \text{Rating Count})$

This formulation balances movie quality and popularity, ensuring that highly rated movies with sufficient user interactions receive higher scores.

4.3 Dynamic Hybrid Weighting

Different users possess varying amounts of rating history. To improve recommendations for cold-start users, dynamic weighting was introduced.

User Type	SVD Weight	Popularity Weight
Cold User (<10 ratings)	75%	25%
Medium User (10–29 ratings)	90%	10%
Heavy User (30+ ratings)	95%	5%

For users with limited interaction history, popularity information contributes more heavily to the final recommendation score. As user activity increases, the system gradually relies more on personalized SVD predictions.

4.4 Candidate Filtering

Movies with very few ratings often produce unreliable recommendation signals. To reduce noise, only movies with at least 20 ratings were considered during recommendation generation.

This filtering strategy improves recommendation reliability while maintaining sufficient recommendation diversity.

4.5 Recommendation Pipeline

The final recommendation process follows these steps:

1. Identify movies not previously rated by the user.
2. Generate predicted ratings using the trained SVD model.
3. Compute popularity scores for candidate movies.
4. Apply dynamic weighting based on user activity level.
5. Rank movies using the final hybrid score.
6. Return the Top-N recommendations to the user.

The final ranked list is displayed through an interactive Streamlit dashboard.

5. Evaluation Metrics

Two metrics were used to evaluate the recommendation system:

5.1 RMSE

RMSE measures rating prediction accuracy by comparing predicted ratings with actual ratings. Lower RMSE values indicate better performance.

5.2 MAP@10

MAP@10 evaluates recommendation ranking quality by measuring how effectively relevant movies appear within the Top-10 recommendations. Higher MAP@10 values indicate better ranking performance.

A movie was considered relevant if its actual rating was greater than or equal to 3.5.

6. Experimental Results

6.1 RMSE Comparison

Model RMSE

SVD	0.961
UserCF	1.028
ItemCF	1.084

Discussion

- SVD achieved the lowest prediction error.
- UserCF performed better than ItemCF but worse than SVD.
- The results indicate that latent factor models are more effective than neighborhood-based approaches for this dataset.

Model Comparison

Recommendation Quality

SVD generated more personalized recommendations by learning hidden user preferences. UserCF and ItemCF relied only on similarity relationships and therefore captured less complex patterns.

Training Complexity

SVD required longer training time due to latent factor optimization, whereas UserCF and ItemCF were comparatively simpler to train.

Computational Efficiency

After training, SVD generated recommendations efficiently using learned user and movie factors. UserCF and ItemCF required similarity computations during recommendation generation.

Practical Usability

SVD provided the best balance between accuracy, scalability, and recommendation quality, making it the most suitable model for deployment.

6.2 MAP@10 Results

Candidate Pool	MAP@10
Min 20	0.0028
Min 50	0.0030
Min 100	0.0036

Discussion

Increasing the minimum rating threshold improved MAP@10 by removing low-signal movies from the candidate pool. However, larger thresholds also reduced recommendation diversity. A threshold of 20 ratings was selected for the final system to maintain a balance between ranking quality and recommendation coverage.

6.3 Final Model Selection

Based on RMSE performance, recommendation quality, computational efficiency, and practical usability, SVD was selected as the foundation of the final recommendation system. The model was further enhanced using dynamic popularity weighting and candidate filtering, resulting in the final Dynamic Hybrid SVD recommender.

7. Recommendation Examples

The final Dynamic Hybrid SVD recommender was evaluated on users with different activity levels to assess recommendation quality under varying interaction histories.

7.1 Heavy User Recommendation Example

User Profile

- User ID: 2439493
- Total Ratings: 140
- User Type: Heavy User
- Weighting Strategy: 95% SVD + 5% Popularity

User History

	movie_id	title	year	rating
402243	17158	The Jungle Book	1994	5
402318	11818	The Bells of St. Mary's	1945	5
402242	5151	Bedknobs and Broomsticks	1971	5
402250	257	Charlotte's Web	1973	4
402246	13301	9/11	2002	4
402207	6887	The Cosby Show: Season 1	1984	3
402204	8640	Little House on the Prairie: The Pilot	2003	3
402214	8733	Doogie Howser, M.D.: Season 1	1989	3
402193	5579	Miami Vice: Season 1	1984	3
402192	15862	A Star Is Born	1954	3

Recommended Movies

	rank	movie_id	title	year	predicted_rating	hybrid_score	ratings_count
0	1	2913	Finding Neverland	2004	3.441	3.474	339
1	2	12034	The Shining	1980	3.103	3.160	382
2	3	17146	Doctor Zhivago	2002	3.085	3.054	23
3	4	1357	Stargate SG-1: Season 7	2003	3.000	3.003	41
4	5	7233	Stand by Me	1986	2.873	2.944	342
5	6	1441	Stargate SG-1: Season 2	1998	2.919	2.928	52
6	7	15296	Band of Brothers	2001	2.819	2.881	136
7	8	14961	Lord of the Rings: The Return of the King: Ext...	2003	2.773	2.876	297
8	9	12834	Family Guy: Vol. 2: Season 3	2001	2.834	2.873	95
9	10	2172	The Simpsons: Season 3	1991	2.812	2.864	111

Analysis

The user's history contains several classic movies and television series, including *The Jungle Book*, *Bedknobs and Broomsticks*, *Little House on the Prairie*, and *Miami Vice*. The recommender generated recommendations such as *Doctor Zhivago*, *Stargate SG-1*, *Band of Brothers*, *The Simpsons*, and *Family Guy*, which also belong to well-established and widely viewed film and television franchises.

The recommendations indicate that the model successfully captured the user's preference for older and long-running content rather than recommending only recently released movies.

Success Factors

- Captured preference for classic and nostalgic content.
- Recommended both movies and television series similar to previously watched titles.
- Leveraged extensive rating history to generate personalized recommendations.

7.2 Cold User Recommendation Example

User Profile

- User ID: 788
- Total Ratings: 4
- User Type: Cold User
- Weighting Strategy: 75% SVD + 25% Popularity

User History

USER HISTORY				
	movie_id	title	year	rating
101	7624	Top Gun	1986	5
100	17324	Hitch	2005	4
102	3371	Whale Rider	2003	4
99	12155	Spider-Man 2	2004	3

Recommended Movies

	rank	movie_id	title	year	predicted_rating	hybrid_score	ratings_count
0	1	11521	Lord of the Rings: The Two Towers	2002	4.969	4.977	523
1	2	7230	The Lord of the Rings: The Fellowship of the R...	2001	4.745	4.753	307
2	3	7193	The Princess Bride	1987	4.769	4.707	368
3	4	6974	The Usual Suspects	1995	4.649	4.663	397
4	5	10042	Raiders of the Lost Ark	1981	4.562	4.650	469
5	6	12870	Schindler's List	1993	4.602	4.647	374
6	7	14961	Lord of the Rings: The Return of the King: Ext...	2003	4.583	4.642	297
7	8	9628	Star Wars: Episode VI: Return of the Jedi	1983	4.636	4.639	336
8	9	7057	Lord of the Rings: The Two Towers: Extended Ed...	2002	4.605	4.633	297
9	10	16954	Indiana Jones and the Last Crusade	1989	4.533	4.605	454

Analysis

The user has very limited interaction history, making preference learning difficult. To address this issue, the system increases the influence of popularity-based ranking.

The recommendations include widely appreciated and frequently rated movies, ensuring stable recommendation quality despite limited user information.

Success Factors

- Handles sparse user profiles effectively.
 - Reduces cold-start impact.
 - Produces reliable recommendations with limited data.
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7.3 Recommendation Quality Assessment

The recommendation examples demonstrate that the Dynamic Hybrid SVD framework adapts its behavior based on user activity levels.

For active users, recommendations are primarily driven by personalized latent factor predictions. For sparse users, popularity weighting improves stability while still preserving personalization.

This adaptive behavior allows the system to maintain recommendation quality across different user categories.

7.4 Failure Cases and Limitations

Although the system performs well in most scenarios, several limitations remain:

- Extremely new users with no rating history cannot receive fully personalized recommendations.
 - Movies with very few ratings may not appear frequently due to candidate filtering.
 - The model relies only on collaborative information and does not utilize genre, cast, or content metadata.
 - Recommendation diversity may decrease for users with very narrow viewing preferences.
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7.5 Key Observations

- Heavy users receive highly personalized recommendations.
- Cold users benefit significantly from popularity weighting.

- Candidate filtering improves recommendation reliability.
 - Dynamic Hybrid SVD adapts effectively to varying user activity levels.
 - The final system balances personalization and recommendation stability.
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8. Key Insights

- The dataset exhibited high sparsity.
 - Approximately 68% of users were cold users.
 - SVD achieved the best RMSE among all evaluated models.
 - Dynamic weighting improved recommendation stability for sparse users.
 - Candidate filtering improved recommendation ranking performance.
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9. Future Improvements

- Incorporate genre and content information.
 - Implement content-based recommendation techniques.
 - Explore Neural Collaborative Filtering models.
 - Support real-time recommendation updates.
 - Incorporate user feedback for continuous model improvement.
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10. Conclusion

This project developed and evaluated multiple collaborative filtering approaches for movie recommendation. SVD achieved the best predictive performance and was selected as the foundation of the final system. By incorporating dynamic popularity weighting and candidate filtering, the final Dynamic Hybrid SVD recommender improved recommendation quality for both active and cold-start users. The deployed Streamlit application demonstrates the practical applicability of the proposed recommendation framework.